



RAMCO INSTITUTE OF TECHNOLOGY

RAJAPALAYAM

Department of Artificial Intelligence and Data Science

Academic Year: 2023- 2024 (ODD Semester)

Active Learning Best practices: Project Based Learning

Degree, Semester & Branch: V Sem. B.Tech. Artificial Intelligence and Data Science

Course Code & Title: - AD3002 –Health Care Analytics

Name of the Faculty member: Dr.S.Vimal ,Associate Professor/AD

Theme of discussion: Disease Identification from a Public/Real data Set and Analysing it using ML/DL algorithm

Date and Time: Oct 12- Oct 31, 2023

Active Learning Best practices: Project Based Learning

Tentative Title of the Project :

1. Patient Risk Prediction
2. Gene Cluster Analysis
3. Pneumonia Detection From X-Rays
4. Cancer Disease Prediction
5. Healthcare Supply Chain Optimization
6. Natural Language Processing for Clinical Notes
7. Liver cirrhosis prediction
8. Multi-sensor Thoracic Data Fusion for Datasets
9. 3D Ultrasound Imaging by Spatial Compound
10. Healthcare Chatbot Development
11. 3D MRI or CT Scan Images Fusion
12. Accurate Clinical Adaptability, Forecasting and Reasoning
13. Memory and Time Efficient 3D Morphological Segmentation
14. 3D Optimal Path Selection using Skeletonization Approach for Brain Blood Vessels
15. Early Tumor Prediction and Diagnosis (Response to Radiation)
16. In homogeneities Correction of MRI Images using InhomoNet

Techniques in addition to be used:

1. Federated Learning
2. Transfer Learning
3. Self-Supervised Learning
4. Multimodal Deep Learning
5. Reinforcement Learning
6. Geometric Deep Learning
7. Distributed Machine Learning
8. Deep Learning Model Complexity
9. Weakly Supervised Learning
10. Incremental Deep Learning
11. Physics-based Machine Learning Model
12. Deep Neural Network (Optimization)
13. Deep Learning
14. Zero-shot Learning

Learning Outcomes:

1. Use Emerging technologies to solve problems and complete projects in their academic, Professional, and personal lives.
2. Apply statistical tools for organizing, analyzing, evaluating, and interpreting data.
3. Apply mathematical analysis to solve problems that they encounter in their academic, professional, and personal lives.

Implementing Project-Based Learning:

- (1) Defining problems in terms of given constraints or challenges
- (2) Generating multiple ideas to solve a given problem,
- (3) Prototyping potential solutions to a problem, and
- (4) Testing the developed solution products

Procedure:

- Each student has been assigned to take a Health Data set from public available source and also from real sources of health data.
- The student has been asked to apply ML/DL algorithms which they have studied in Laboratory.
- Implement deep learning models in Python using the PyTorch and TensorFlow libraries and train them with real-world datasets. Also visualise them using any DEV Tools
- The Chosen data Set has been trained and implemented using any AI tools with the techniques given
- They are asked to check the accuracy of the identified disease and finally that has to be posted in their repository
- The student Github /LinkedIn pages with their project has been verified with the project development Rubrics (Design Methodology, Planning of Project Work and Team Structure, Demonstration and Presentation)
- The students were asked to give a short presentation on what they learned in the project and how the future problems can be addressed and what are the challenges they have faced while handling the dataset

At the end of the project submission, student shared the knowledge what they learnt as a presentation

Glimpses:

Video Source Link of Presentation:

https://drive.google.com/file/d/1p4x01UlogZP39gd_P9yJzP7vuCyOdfjU/view?usp=drive_link

https://drive.google.com/file/d/1NixBsig7SncgqMn1K6-6Mf5m1bar6Jwx/view?usp=drive_link

https://drive.google.com/file/d/1Vq9l43QpldaffWnmmQ4mWCv_fBTfbbbD/view?usp=drive_link

Project Details:

https://docs.google.com/spreadsheets/d/1CJkP-hZqWWg3mIKfc-EiJ1hbGYsknr9Q/edit?usp=drive_link&ouid=117962305416782579351&rtpof=true&sd=true

Some Screenshots of Presentation:



Comments on Student Deliverables of Project in Social Media pages:

**THIRUNIRAISELVI M** • 1st
AI student
3w •

Exciting Journey into Gene Expression Analysis! 📊

I recently completed a fascinating project in gene expression analysis, and I'm thrilled to share the highlights of my work. In this project, I explored the intricate world of genomics, transforming raw data into valuable insights.

Here's a glimpse into what I accomplished:

- 🔧 **Data Preparation:** I started by importing the necessary libraries and the dataset. After cleaning and transforming the data, I preprocessed it for analysis.
- 📊 **PCA for Dimensionality Reduction:** Principal Component Analysis (PCA) helped reduce the dataset's dimensionality while retaining essential information. I visualized the data in 2D and 3D, revealing distinct patterns.
- 🏗️ **Model Building:** I applied various machine learning algorithms to classify acute lymphoblastic leukemia (ALL) and acute myeloid leukemia (AML) based on gene expression profiles. Here are some key highlights:

K-Means Clustering
Naive Bayes
Logistic Regression
Support Vector Machine
Random Forest
XGBoost

**Max Jonas Gripa** (He/Him) • 3rd +
Full-Stack Web Developer | Next.js | Node | PostgreSQL | Google Cloud |...
2w •
Awesome and inspiring job. Congrats! 🎉👏

Like | Reply

**Marees Waran** • 1st
Pre Final Year Student at RAMCO INSTITUTE OF TECHNOLOGY
3w •

Exciting News: Successfully Completed a Healthcare Chatbot Development Mini Project in healthcare analytics! 📊

I'm thrilled to share that I've just wrapped up a rewarding mini project in healthcare chatbot development! 🎉

🔍 **Project Overview:**
For the past few weeks, I've been working on creating an intelligent and user-friendly healthcare chatbot to assist patients and healthcare professionals in accessing vital information and support.

🎯 **Project Goals:**

Enhancing Patient Experience: The primary goal was to develop a chatbot that could provide quick, accurate, and empathetic responses to patient inquiries, ranging from appointment scheduling to general health information.

Efficient Information Retrieval: We aimed to streamline the process of accessing medical knowledge and resources for healthcare practitioners, making their jobs more efficient.

Data Privacy & Security: A significant focus was on ensuring the chatbot adhered to strict privacy and security standards, guaranteeing the protection of sensitive healthcare data.

github link : <https://lnkd.in/g/fjM5nyh>

**Dr. Sushil Kumar Singh** • 1st
Phd (SeoulTech) | IEEE Member | Researcher | Guest Lecturer | Associa...
2w •
Nice

Like | 1 | Reply | 1 Reply

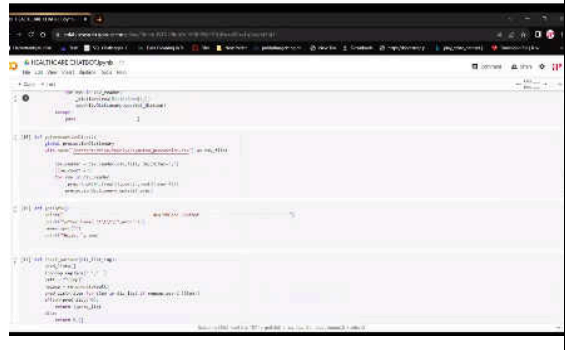
**Marees Waran** (He/Him) **Author**
Pre Final Year Student at RAMCO INSTITUTE OF TECHNOLO...
2w •
Thank you sir.

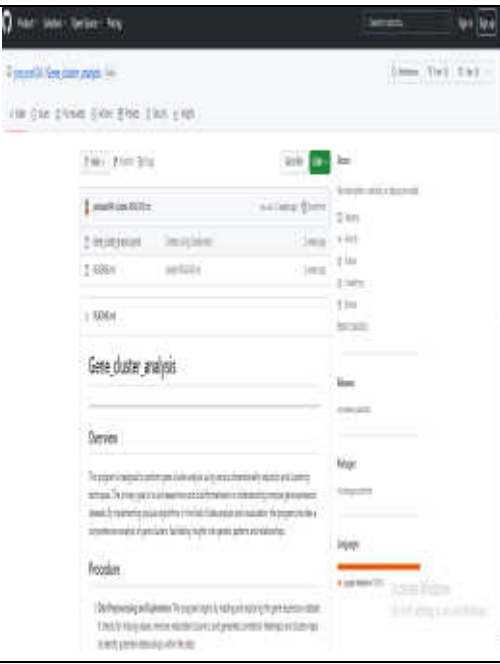
1 Like | 1 Danku

**Pabintha A** • 2nd
Data Analyst
3w •

Excited to share my new Healthcare Chatbot! 📊 It's a friendly tool that helps us to understand our symptoms and suggests possible health issues. With this chatbot, we aim to empower users to take proactive steps towards better well-being. Leveraging advanced algorithms, including Decision Tree Classifier and Support Vector Machine, the chatbot offers personalized health insights based on the symptoms."

#HealthTech #AI #HealthcareInnovation
#Healthcare #Chatbot #Machinelearning





Reflective Report:**Challenges and strategies suggested by students:**

- To motivate the slow learners to involve the activity.
- Inference collection about the previous done project with respective to the chosen disease
- Collecting Real Time dataset for Genomic based disease prediction
- Making the students inline with the timely submission
- Handling the huge voluminous data set , since it needs more disk space for Training and validating the data set
- As Individual project has been planned , students take more timelines to deliver the completed project

Observations:

Using Project based Learning is one of the most common study methods it help students to engage them to work in a implementation of their ideas towards a given problem. Some Students learnt new tools for implementation of their chosen health data set that creates a impact for other students to innovate their project ideas with emerging tools like IBM Cognos Analytics.

Students Response:

- Bright students were actively participated to use advanced DL algorithms
- Slow learners expect to assist their project implementation and they have successfully completed the assigned task
- Mentor has involved for successful execution and validation of results
- Most of the students have actively done and posted their findings in their Github and other repositories



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FEEDBACK

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ML/DL algorithm

Date and Time: Oct 12- Oct 31, 2023

Feedback collected in class and also through online

Gform Link: <https://forms.gle/EHAjM7Hw6Yi1T2cA9>

Feedback Questionnaire:

1.

Feed back- Project Based Learning

1. Email *

2. 1. Does the Mini project assigned gained insights of the latest technologies learned in Health care Analytics

Mark only one oval.

☐ Yes

☐ No

3. 2. Can you able to prepare a EHR for the given medical data set.

Mark only one oval.

☐ Yes

☐ No

4. 3. Have you done the project individually or Group

Mark only one oval

Activate Windows
Go to PC settings to activate Windows.

☐ Yes
☐ No

3. 2. Can you able to prepare a EHR for the given medical data set.
Mark only one oval.
☐ Yes
☐ No

4. 3. Have you done the project individually or Group
Mark only one oval.
☐ Individual
☐ Group

5. 4. Have you applied the ML/DL algorithms for the development of the project
Mark only one oval.
☐ Yes

Activate Windows
Go to PC settings to activate Windows.

feedback questionnaire.pdf - Reader

2 4

7. 6. Does the Mini project and project based Learning helps you to explore the ideas that you can create for a future clinical decision making.
Mark only one oval.
☐ Yes
☐ No

8. 7. Rate your experience on working with the project and also the outcome analysis
Mark only one oval.
☐ Excellent
☐ Good
☐ Fair

9. 8. Any other comments

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Feedback Analysis:

Feed back- Project Based Learning

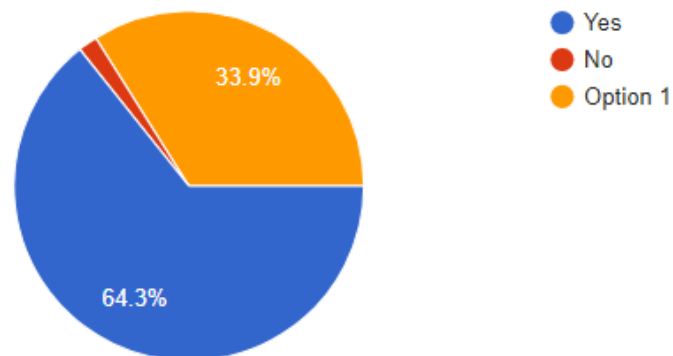
56 responses

[Publish analytics](#)

1. Does the Mini project assigned gained insights of the latest technologies learned in Health care Analytics

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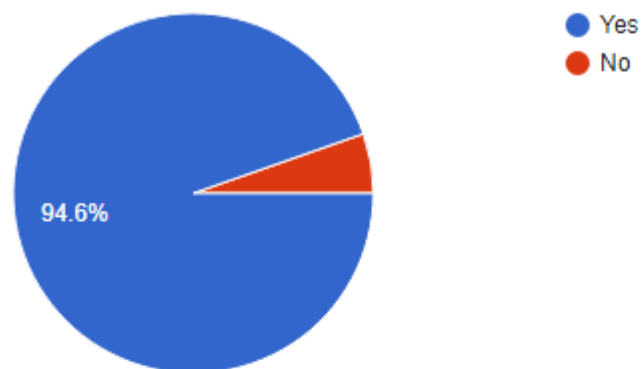
56 responses



2. Can you able to prepare a EHR for the given medical data set.

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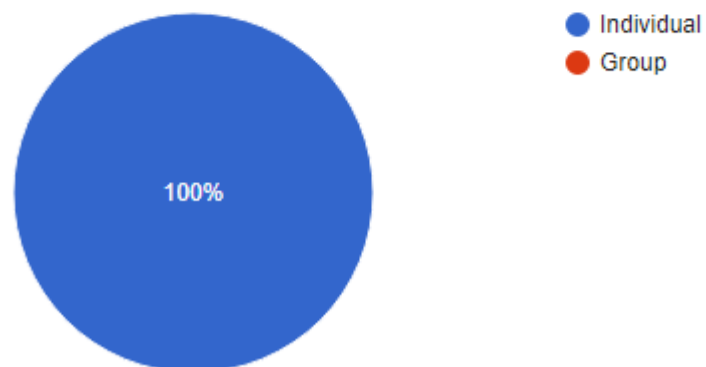
56 responses



3. Have you done the project individually or Group

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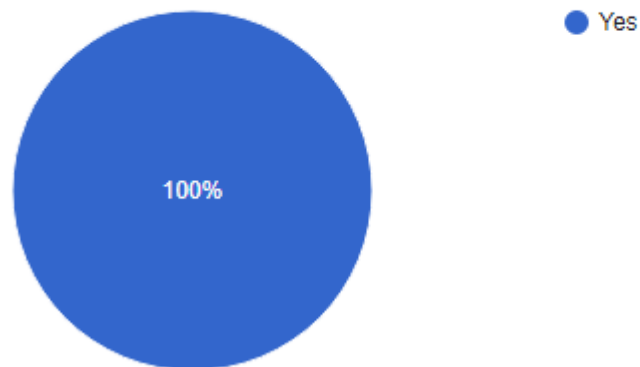
56 responses



4. Have you applied the ML/DL algorithms for the development of the project

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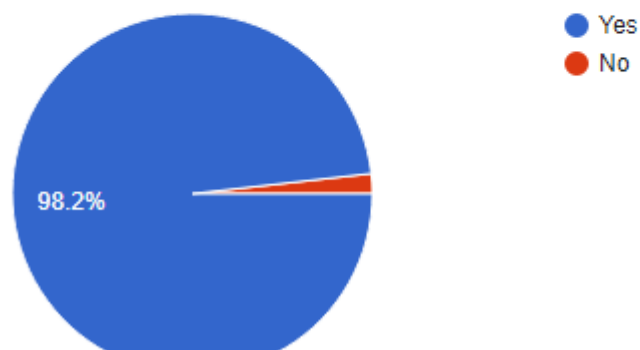
55 responses



5. Does the outcome from your project meets your inference obtained from previous existing project

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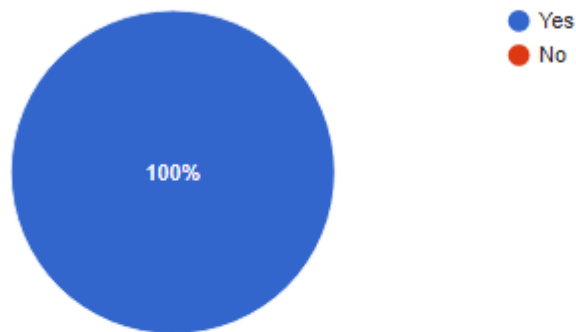
56 responses



6. Does the Mini project and project based Learning helps you to explore the ideas that you can create for a future clinical decision making.

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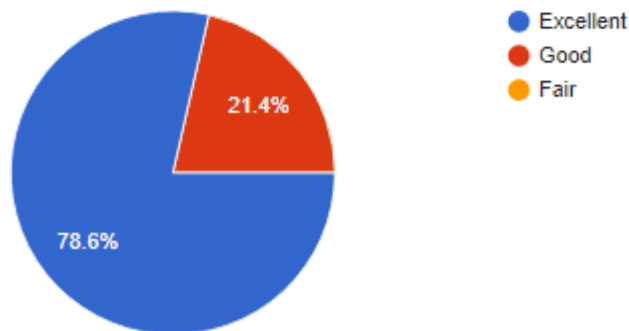
56 responses



7. Rate your experience on working with the project and also the outcome analysis

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56 responses



5. Any other comments

37 responses

Nothing

null

Null

NONE

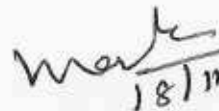
Learnt a new tool IBM Cognos Analytics for visualization

null

Null


(8/11/23)

Course In charge
[Dr.S.Vimal, ASP/AD]


18/11/23

HOD/AD
Dr.M.Kaliappan